

RESEARCH

Open Access



Groundwater potential and recharge zoning in semi-arid Deccan basalt using AHP, infiltration testing and VES

Taufique Warsi^{1,2*} , Siddhant Sanjay Sonde^{2,3} and George Biswas^{4*}

*Correspondence:

Taufique Warsi

taufiquewarsi@gmail.com

George Biswas

george.geol@presiuniv.ac.in

¹Raintree Foundation, Wagle Industrial Estate, Rd Number 16, Vaity Wadi, Thane West, Thane, Maharashtra 400604, India

²WOTR Centre for Resilience Studies (W-CReS), Watershed Organisation Trust (WOTR), Pune, Maharashtra, 411009, India

³Department of Geography (Geoinformatics), Savitribai Phule Pune University, Pune, Maharashtra 411007, India

⁴Department of Geology, Presidency University, 86/1 College Street, Kolkata 700073, India

Abstract

Groundwater targeting in semi-arid basaltic terrains is uncertain because surface indicators of recharge, storage and transmissivity do not necessarily represent the same hydrogeological process. This study develops a field-constrained workflow for delineating groundwater potential zones and recharge-prone zones in the Moha-Karewadi area of the Deccan Traps, Beed district, Maharashtra, India. Seven surface predictors - drainage density, lineament density, slope, land use/land cover, lithology, soil and geomorphology - were prepared in a GIS environment and weighted by the Analytical Hierarchy Process (AHP) to generate the Groundwater Potential Zone (GWPZ) map. A second AHP model incorporated a measured infiltration-rate layer derived from Moha-Karewadi site-level field records to generate the Groundwater Recharge Zone (GWRZ) map; external records from other sites were retained only for data audit and were not used as validation of the final map. Subsurface constraints were provided by eight Schlumberger Vertical Electrical Soundings (VES), interpreted as one-dimensional layered-earth models and assembled into pseudo-sections and geoelectric sections. Very high groundwater potential is spatially restricted, covering about 2.07% of the area, whereas very high recharge suitability covers about 8.86% and occurs mainly where favourable geomorphology, low drainage density, high lineament density and higher infiltration capacity coincide. VES interpretation indicates shallow weathered and fractured basalt near the tanks, commonly within about 7–10 m below ground level, underlain locally by compact basalt that restricts deeper percolation. The main contribution is the separation of groundwater potential and recharge suitability as related but non-equivalent decisions, followed by reconciliation with infiltration and resistivity evidence. The workflow provides a practical screening approach for managed aquifer recharge and groundwater exploration in hard-rock watersheds.

Keywords Groundwater potential zone, Groundwater recharge zone, Analytical hierarchy process, Vertical electrical sounding, Infiltration rate



1 Introduction

Groundwater in hard-rock and basaltic terrains is commonly stored and transmitted through secondary porosity, including weathered mantle, joints, fractures, vesicles and interflow zones. In such aquifers, productive zones are spatially discontinuous, and borewell success depends on the coincidence of recharge, storage and transmissive pathways rather than on lithology alone [26, 28, 29, 40]. These controls are particularly important in semi-arid regions where monsoon rainfall is concentrated in a short season and groundwater abstraction supports domestic water supply and agriculture.

Remote sensing and GIS-based multicriteria methods, especially AHP and related MCDA approaches, are widely used for groundwater potential mapping because they can combine geomorphology, drainage, lineament, slope, soil, geology and land-use information over large or data-scarce areas [18, 23, 33]. Applications from crystalline, deltaic, volcanic and semi-arid terrains demonstrate the utility of such approaches, including geophysical-topographic integration, fuzzy-AHP, watershed-scale groundwater-potential zoning and recharge-priority mapping [1, 2, 4, 8, 14, 16, 20, 22, 35]. However, these studies also highlight an important limitation: surface-derived suitability does not automatically confirm the presence of a productive or rechargeable aquifer at depth [10, 21].

This limitation is critical in the Deccan Traps, where weathered basalt may favour shallow storage, compact basalt may impede vertical percolation, and fractured or vesicular units may provide localised pathways. A location may therefore appear favourable in a surface-based groundwater potential map but still be unsuitable for artificial recharge if near-surface infiltration is low or if compact basalt restricts downward movement. Comparable groundwater quality, vulnerability, managed-aquifer-recharge and prediction studies show that mapped indicators must be tested against hydrological or aquifer evidence before management decisions are made [3, 5, 19, 34, 36]. This distinction explains why groundwater-potential zoning and recharge zoning should be treated as related but different decisions.

Previous studies have combined geospatial methods with geophysical surveys or hydrogeological indices to improve groundwater targeting [6, 21, 30–32, 37, 38]. Nevertheless, many local applications still present AHP-derived maps descriptively, with limited explanation of why layers were selected, how weights were validated, how infiltration modifies recharge suitability, and how resistivity sections support or constrain the final interpretation. These gaps reduce the reproducibility and field value of the maps.

The present study addresses this gap through a field-constrained assessment of the Moha-Karewadi watershed, where percolation tanks and recharge interventions require spatial prioritisation. The working hypothesis is that a single surface-derived suitability map cannot adequately represent both groundwater occurrence and recharge acceptance in basaltic terrain. Accordingly, the study integrates (i) a seven-layer AHP model for Groundwater Potential Zones, (ii) a separate recharge model that includes measured infiltration rates, and (iii) VES-based subsurface validation. The objectives are to delineate groundwater potential zones, identify recharge-prone zones, test the hydrogeological consistency of both outputs against field evidence, and develop a transparent workflow for managed aquifer recharge planning in comparable hard-rock watersheds.

2 Study area

The study area covers the adjoining villages of Moha and Karewadi in Manjalgaon taluka, Beed district, Maharashtra, India. The mapped area is approximately 32 km² and lies within the semi-arid Deccan basalt terrain of the Marathwada region. The work focused on the surroundings of percolation tanks, including a tank desilted during 2016–2017, because these structures are locally important for recharge and water security. The area lies near 18°52′16.104″ N, 76°17′56.004″ E to 18°52′56.3988″ N, 76°18′59.1264″ E. The region experiences hot and dry pre-monsoon conditions, with peak summer temperatures approaching 42 °C, and receives most rainfall during the June–September monsoon. Average monsoon rainfall is about 737 mm. The terrain includes dissected structural lower plateau, pediment-pediplain complexes, water bodies and black cotton soils. These features make the area suitable for testing whether surface-based groundwater and recharge maps can be strengthened by infiltration and VES evidence. The location and elevation setting of the study area are shown in Fig. 1.

2.1 Geological and hydrogeological setting

The Moha-Karewadi area forms part of the Deccan Volcanic Province and is underlain by Late Cretaceous-Palaeocene basalt flows. The lithological map used in this study distinguishes six basalt map units (codes 0832–0837); at the scale of this investigation, however, groundwater behaviour depends more directly on flow facies, weathering, joints and fracture continuity than on the map-unit label alone. Regionally, Deccan basalt successions comprise alternating massive or compact flow interiors and vesicular-amygdaloidal, brecciated or weathered flow-top zones. Interflow contacts and weathered or fractured horizons can provide secondary porosity and transmissivity, whereas unfractured massive basalt and clay-rich interflow material may restrict vertical percolation [15, 27].

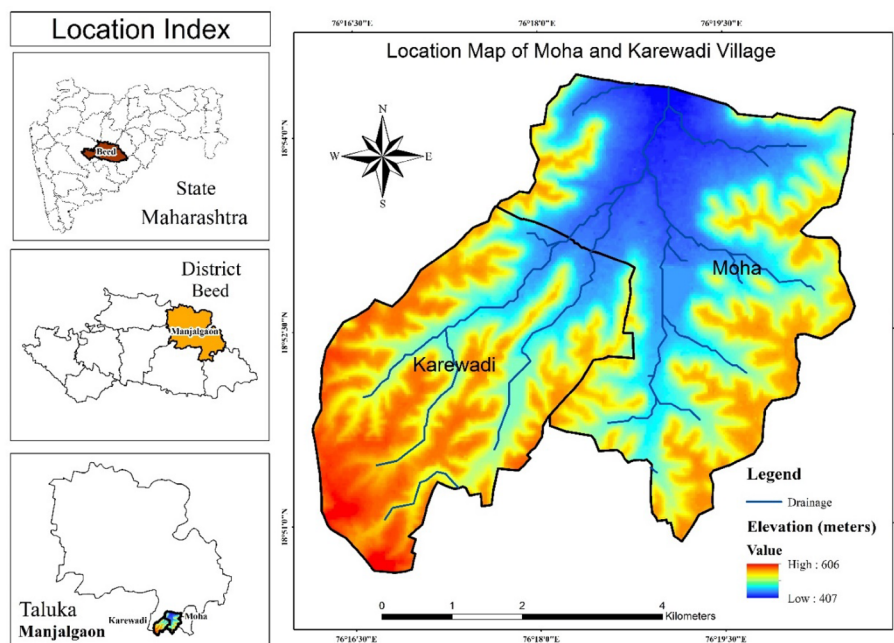


Fig. 1 Location map of the Moha-Karewadi study area in Beed district, Maharashtra, India

Within the study area, highly dissected structural lower plateaus dominate the surface setting, with local pediment-pediplain surfaces, ponds and water bodies. Black cotton soils overlie the basalt; their clay-rich character can favour near-surface storage but may suppress intake where compaction, swelling or waterlogging occurs. The working hydrogeological model is therefore a discontinuous shallow weathered/fractured basalt aquifer, with local deeper fracture or vesicular pathways and comparatively resistive compact-basalt intervals that constrain downward movement. This geological framework provides the basis for including geomorphology and lineament density in the AHP, using infiltration tests to assess near-surface acceptance, and applying VES to distinguish likely weathered/fractured zones from compact basalt [7, 39]. The model is evaluated against the field evidence presented in Sect. 4.9.

3 Materials and methods

The methodological sequence was designed to keep the potential-zone and recharge-zone decisions separate before field-based validation. The overall workflow is shown in Fig. 2.

3.1 Data, layers and analytical components

The workflow separates surface predictors, near-surface infiltration measurements and subsurface validation rather than merging all evidence into a single map overlay. Table 1 summarises the role of each analytical component in the groundwater-potential and recharge-zoning framework.

3.2 Thematic layers and rationale for selection

The seven thematic layers were selected because each represents a first-order control on groundwater occurrence in hard-rock terrain. Drainage density acts as an inverse indicator of infiltration opportunity and residence time; lineament density represents fracture-controlled permeability; slope controls runoff versus infiltration; land use/land cover controls surface roughness, water retention and imperviousness; lithology represents the hydrostratigraphic framework; soil affects near-surface storage and infiltration; and

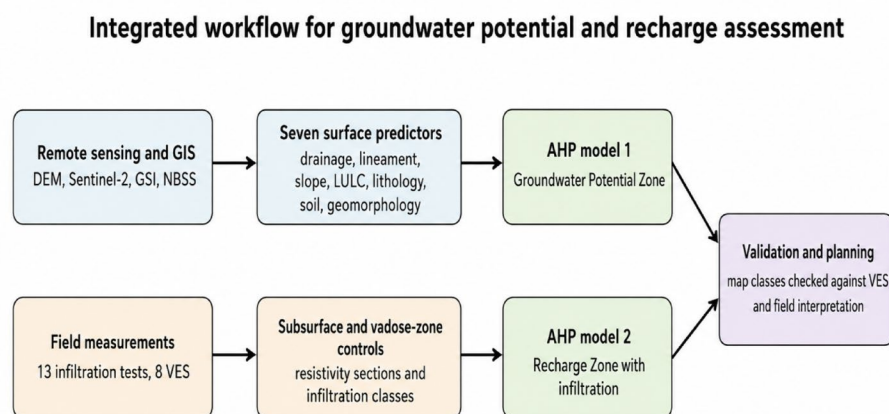


Fig. 2 The workflow showing the separation between groundwater potential zoning and recharge zoning, followed by field validation

Table 1 Compact methods summary for the integrated workflow

Component	Input data/field data	Processing/analysis	Purpose in the workflow
AHP-based Groundwater Potential Zone (GWPZ)	Drainage density, lineament density, slope, land use/land cover, lithology, soil and geomorphology	Reclassification of each thematic layer; pairwise comparison using Saaty scale; normalised weights; weighted overlay	Identifies zones where surface indicators favour groundwater occurrence and storage.
AHP-based Groundwater Recharge Zone (GWRZ)	The seven GWPZ layers plus field-measured infiltration-rate layer	Separate AHP weighting with infiltration added as an independent near-surface recharge control	Separates recharge suitability from groundwater potential, because high storage potential does not always imply high infiltration acceptance.
Double-ring infiltration test	Moha-Karewadi site-level infiltration records; external Beed/Nanded records retained only for data audit	Constant-head double-ring measurements for 2–4 h; near-steady infiltration rate used for class ranking and interpolation	Adds measured vadose-zone evidence and constrains recharge-prone areas without treating external audit records as validation points.
Vertical Electrical Sounding (VES)	Eight Schlumberger VES points along accessible downstream/upstream traverses near Moha and Karewadi tanks	Apparent resistivity calculation; 1D layered-earth inversion in IPI2DWin; pseudo-sections and geoelectric sections	Checks whether mapped favourable zones coincide with weathered/fractured basalt rather than compact massive basalt.
Validation synthesis	GWPZ/GWRZ maps, infiltration classes and VES interpretations	Overlay comparison and hydro-geological consistency check; class-wise quantitative summary from mapped outputs	Provides field support for the final zoning and identifies where intervention is appropriate.

geomorphology indicates erosion, depositional surfaces and zones of possible weathered mantle development. These layers were used for the GWPZ model. The infiltration-rate layer was not included in the GWPZ model because recharge suitability and groundwater potential are not identical; it was added only in the GWRZ model to capture measured near-surface acceptance.

3.3 Analytical hierarchy process and weighted overlay

AHP was used to convert expert hydrogeological judgement into reproducible weights following Saaty's pairwise-comparison approach [24, 25]. The reciprocal comparison matrix in Eq. (1) compares criterion i with criterion j using the 1–9 importance scale; the normalised principal eigenvector provides the criterion weight w_i . Although other MCDA formulations are available for water-resource decision problems [11], AHP was selected because it allows transparent layer weighting, class ranking and consistency checking.

$$A = [a_{ij}], \quad a_{ij} = \frac{1}{a_{ji}}, \quad a_{ii} = 1, \quad \sum_{i=1}^n w_i = 1 \quad (1)$$

$$CI = \frac{\lambda_{\max} - n}{n - 1}, \quad CR = \frac{CI}{RI} \quad (2)$$

Here CI is the consistency index, $\lambda_{\max}$ is the maximum eigenvalue of the comparison matrix, n is the number of criteria and RI is the random index. Only matrices with $CR \leq 0.10$, calculated using Eq. (2), were accepted for weighted overlay. The GWPZ and GWRZ indices were then calculated using Eq. (3) and Eq. (4), respectively. Criterion weights and detailed class ranks are reported in Table 2 to make the overlay reproducible.

$$GWPZI = \sum_{i=1}^n w_i r_i \quad (3)$$

Table 2 AHP weights and class ranks used for Groundwater Potential Zone (GWPZ) and Groundwater Recharge Zone (GWRZ) mapping

Layer	Weight without infiltration (%)	Weight with infiltration (%)	No. of classes	Class description	Rank
Geomorphology	38	30	4	Highly dissected structural lower plateau	2
Geomorphology	38	30	4	Pediment-pediplain complex	9
Geomorphology	38	30	4	Ponds	5
Geomorphology	38	30	4	Water bodies	7
Drainage density	19	16	5	Very low	9
Drainage density	19	16	5	Low	7
Drainage density	19	16	5	Moderate	3
Drainage density	19	16	5	High	2
Drainage density	19	16	5	Very high	2
Lineament density	12	10	5	Very low	2
Lineament density	12	10	5	Low	2
Lineament density	12	10	5	Moderate	3
Lineament density	12	10	5	High	7
Lineament density	12	10	5	Very high	9
LULC	10	10	6	Water bodies	7
LULC	10	10	6	Irrigation	9
LULC	10	10	6	Fallow land	3
LULC	10	10	6	Settlement	5
LULC	10	10	6	Wasteland	2
LULC	10	10	6	Vegetation	1
Lithology	8	8	6	Late Cretaceous-Palaeocene basalt 0832	7
Lithology	8	8	6	Late Cretaceous-Palaeocene basalt 0833	1
Lithology	8	8	6	Late Cretaceous-Palaeocene basalt 0834	2
Lithology	8	8	6	Late Cretaceous-Palaeocene basalt 0836	2
Lithology	8	8	6	Late Cretaceous-Palaeocene basalt 0837	1
Lithology	8	8	6	Late Cretaceous-Palaeocene basalt 0835	9
Slope	7	7	5	0–4 degrees	9
Slope	7	7	5	4–8 degrees	7
Slope	7	7	5	8–12 degrees	3
Slope	7	7	5	12–16 degrees	2
Slope	7	7	5	> 16 degrees	2
Soil	6	6	2	Calcareous clayey soil	5
Soil	6	6	2	Clayey soil	3
Infiltration	–	13	5	4.01–28.68 mm/h	1
Infiltration	–	13	5	28.68–47.87 mm/h	3
Infiltration	–	13	5	47.87–68.43 mm/h	5
Infiltration	–	13	5	68.43–97.90 mm/h	7
Infiltration	–	13	5	97.90–178.77 mm/h	9

$$\text{GWRZI} = \sum_{i=1}^m w'_i r'_i \quad (4)$$

In Eq. (4), the primed weights and ranks refer to the recharge-zone model, in which infiltration rate is treated as an additional near-surface recharge-control layer.

3.4 Infiltration-rate measurement

Available field infiltration records were screened before use. Only the Moha-Karewadi site-level records located directly within the mapped study area were used to support the recharge interpretation; additional Beed and Nanded records in the raw sheets were

retained only for data audit and were not used as independent validation of the Moha-Karewadi maps. At each included site, the inner and outer rings were maintained at an equal water head of about 10–15 cm to reduce lateral flow from the inner ring. Measurements continued for 2–4 h, and the near-steady infiltration rate was used for classification. The audited Moha-Karewadi near-steady values span approximately 8.41–168.88 mm/h across tank-bed and downstream/peripheral settings, reflecting soil texture, clay content, silt accumulation, geomorphic position and surface condition. The interpolation used for the recharge map was classified into five infiltration-rate classes from 4.01 to 178.77 mm/h. The field arrangement and representative decline of infiltration rate toward a steady state are shown in Fig. 3. The interpretation is consistent with infiltration and near-surface hydrologic studies showing that soil structure, antecedent condition, drainage and surface disturbance strongly affect percolation behaviour [9, 12, 13, 17, 41].

3.5 Vertical electrical sounding and pseudo-section interpretation

VES was carried out using the Schlumberger electrode configuration. Current was injected through electrodes A and B, and potential difference was measured between M and N. The apparent resistivity was calculated using Eq. (5):

$$\rho_a = K \left(\frac{\Delta V}{I} \right), \quad K = \frac{\pi \left[(AB/2)^2 - (MN/2)^2 \right]}{MN} \quad (5)$$

where ρ_a is apparent resistivity, K is the geometric factor, ΔV is potential difference and I is current. A pseudo-section is a graphical arrangement of apparent resistivity values against electrode spacing and traverse position. It is not a direct geological cross-section; rather, it helps visualise vertical and lateral resistivity variation before or alongside inversion. IPI2DWin was used for 1D layered-earth interpretation by forward modelling the

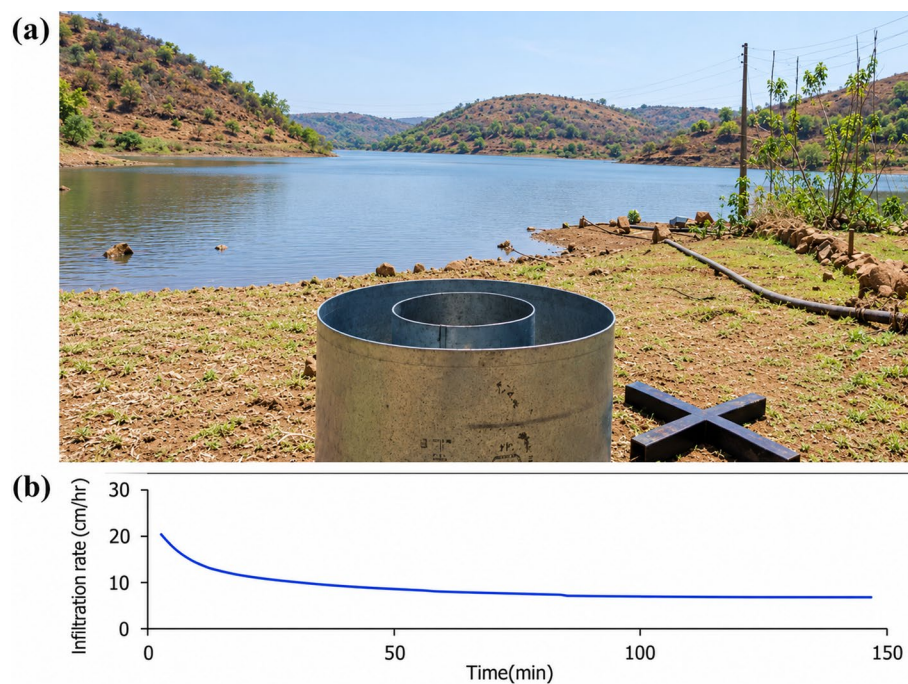


Fig. 3 Field double-ring infiltrometer and typical decline of infiltration rate toward a steady state

apparent-resistivity curve and iteratively adjusting layer resistivity and thickness until the calculated curve matched the measured curve within an acceptable misfit. Lithological interpretation was made by comparing resistivity magnitude, curve shape, depth continuity and field geological context. Low to moderate resistivity at shallow depths was interpreted as weathered or fractured basalt where supported by setting and curve behaviour; relatively high and laterally persistent resistivity was interpreted as compact or massive basalt. The interpretation was therefore hydrogeological rather than purely colour-based.

Recent Deccan basalt resistivity studies also show that conductive or moderate-resistivity anomalies should be interpreted together with geological setting and field evidence rather than treated as direct proof of aquifer productivity [6].

3.6 Statistical and validation analysis

Quantitative evaluation used four complementary checks: class-wise area statistics for the GWPZ and GWRZ maps, hydrogeological consistency between mapped classes and VES-derived lithological interpretation, an audit of the raw infiltrometer data sheets, and an ANOVA-ready model for future independent grouped testing. The raw infiltrometer sheets contain repeated time-step readings within individual tests. These readings describe the decline of infiltration rate toward a near-steady condition and were therefore not treated as independent spatial replicates. For site-level screening, the near-steady infiltration metric was defined as the mean of the final three valid rate readings for a test, where at least three readings were available. One-way ANOVA is appropriate only when these independent site-level values are grouped by a predefined soil, geomorphic or recharge class, as indicated in Eq. (6).

$$H_0 : \mu_1 = \mu_2 = \dots = \mu_k; F = \frac{MS_{\text{between}}}{MS_{\text{within}}} \quad (6)$$

The available records support descriptive site-level screening, but they do not provide a complete georeferenced table in which every independent infiltration record is assigned to the final soil, geomorphic and recharge-map classes. Inferential p values were therefore not used as evidence for the final zoning, because doing so would either rely on pseudo-replication of time-step readings or on incomplete class membership. The analysis instead reports reproducible class-area statistics, VES-based hydrogeological validation and the ANOVA-ready model. The data-screening outcome is provided in Supplementary File S1.

This treatment follows the physical-hydrology requirement that infiltration must be interpreted from independent field tests and site context, not from repeated time-step observations alone [9]. It also keeps the statistical interpretation conservative: the map is validated by physically independent evidence from VES and field interpretation, while the ANOVA framework is supplied for future datasets with replicated site-wise class assignments.

4 Results and discussion

4.1 Drainage density

Drainage density is inversely related to infiltration opportunity in this setting. Areas of very low to low drainage density favour longer residence time and greater percolation,

whereas high drainage density indicates greater surface runoff and limited infiltration. The drainage-density classes were derived from a 30 m DEM using sink filling, flow direction and flow accumulation routines. The 0–0.5 km/km² class was ranked most favourable, and areas above about 2.39 km/km² were ranked least favourable. The mapped drainage-density distribution is shown in Fig. 4.

4.2 Lineament density

Lineaments represent potential fractures, joints and structural pathways that can enhance groundwater movement in basaltic terrain. Higher lineament density was therefore assigned higher rank in the GWPZ model. In the study area, lineament densities of about 2.1–6.9 km/km² were treated as favourable, while 0–1.3 km/km² was treated as low-favourability. The discussion clarifies that lineament density is a proxy and must be interpreted with VES and geomorphology rather than as direct proof of aquifer productivity. The lineament-density map is shown in Fig. 5.

4.3 Slope

Slope influences the partitioning of rainfall into runoff and infiltration. Lower slopes promote ponding, longer residence time and greater recharge opportunity, whereas steep dissected slopes promote rapid runoff. The 0–4-degree class was therefore ranked highest, while slopes above 16 degrees were ranked lowest. The slope classification is shown in Fig. 6.

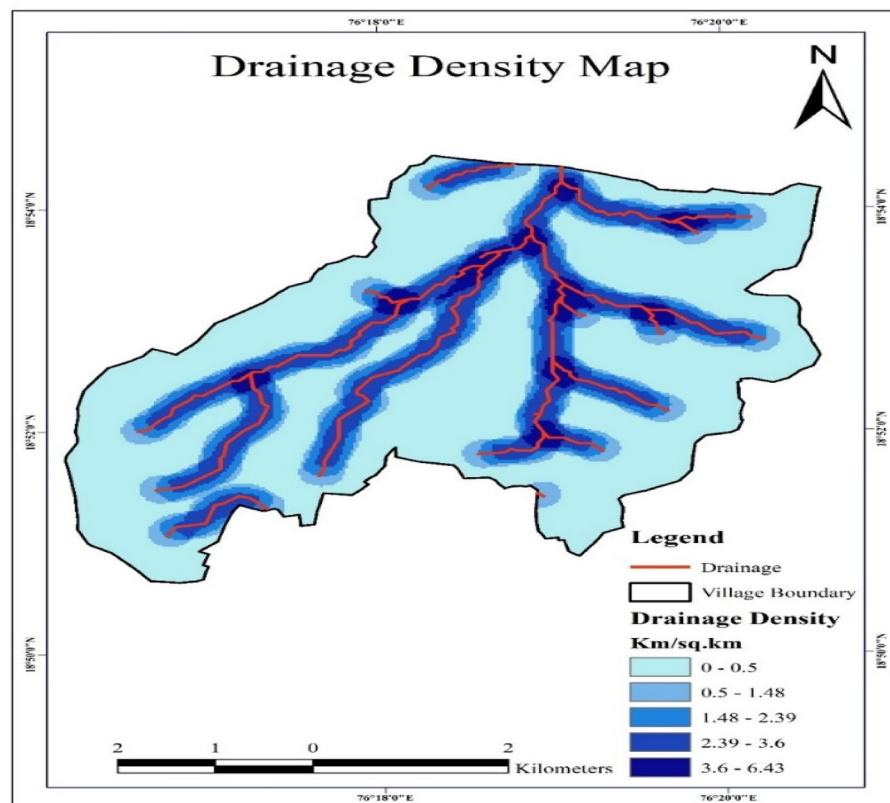


Fig. 4 Drainage density map of the study area

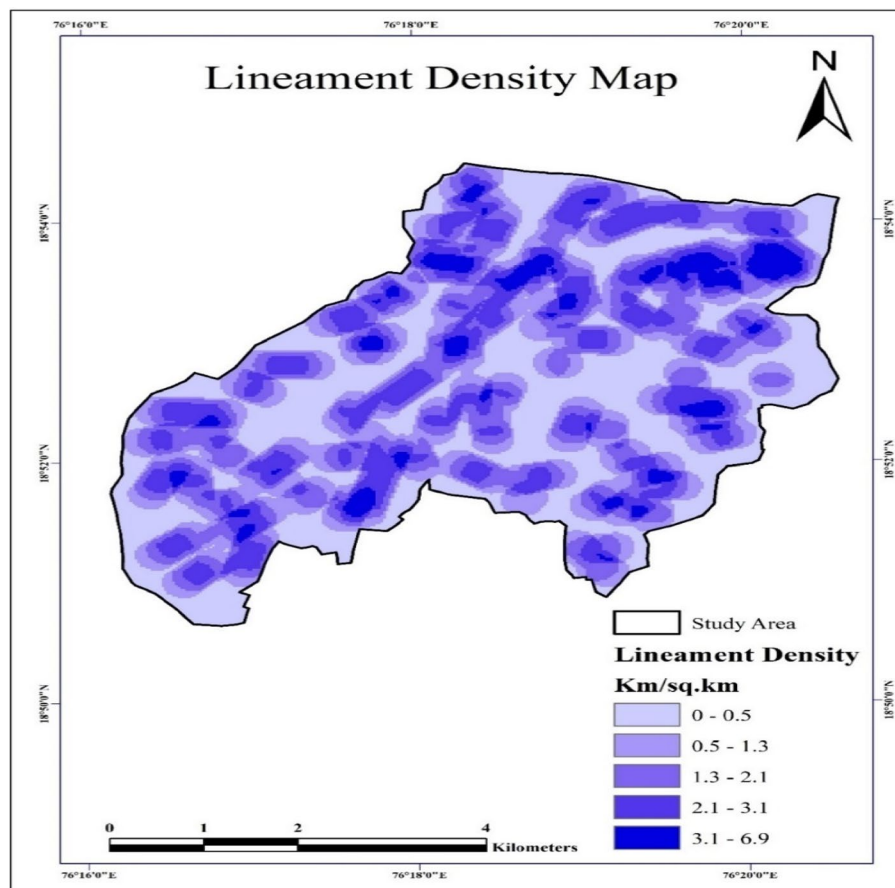


Fig. 5 Lineament density map derived from DEM-based hillshade interpretation

4.4 Geomorphology

Geomorphology provides an integrated expression of erosion, deposition, topographic position and likely weathered-zone development. Highly dissected structural lower plateau occupies about 2530.42 ha and was ranked low because it promotes runoff and discontinuous storage. Pediment-pediplain complexes, ponds and water bodies were ranked higher because they favour water retention, sediment accumulation and recharge opportunity. The spatial distribution of geomorphic classes is shown in Fig. 7.

4.5 Soil

The soil map shows calcareous clayey and clayey soils. Clayey soils can store water but may reduce infiltration when swelling, compaction or silt accumulation is high. Therefore, soil was not interpreted alone. In pediment-pediplain settings, calcareous clayey soil may support moderate infiltration where slope is low and the vadose zone is not clogged by fine sediment. In steeper dissected areas, the same soil texture is less favourable because runoff dominates. The soil distribution is shown in Fig. 8.

4.6 Lithology

Lithology controls the hydrostratigraphic framework, but the groundwater significance of individual basalt units depends on weathering, fractures, vesicularity and topographic position. Low-lying basalt units associated with pediment-pediplain areas were assigned

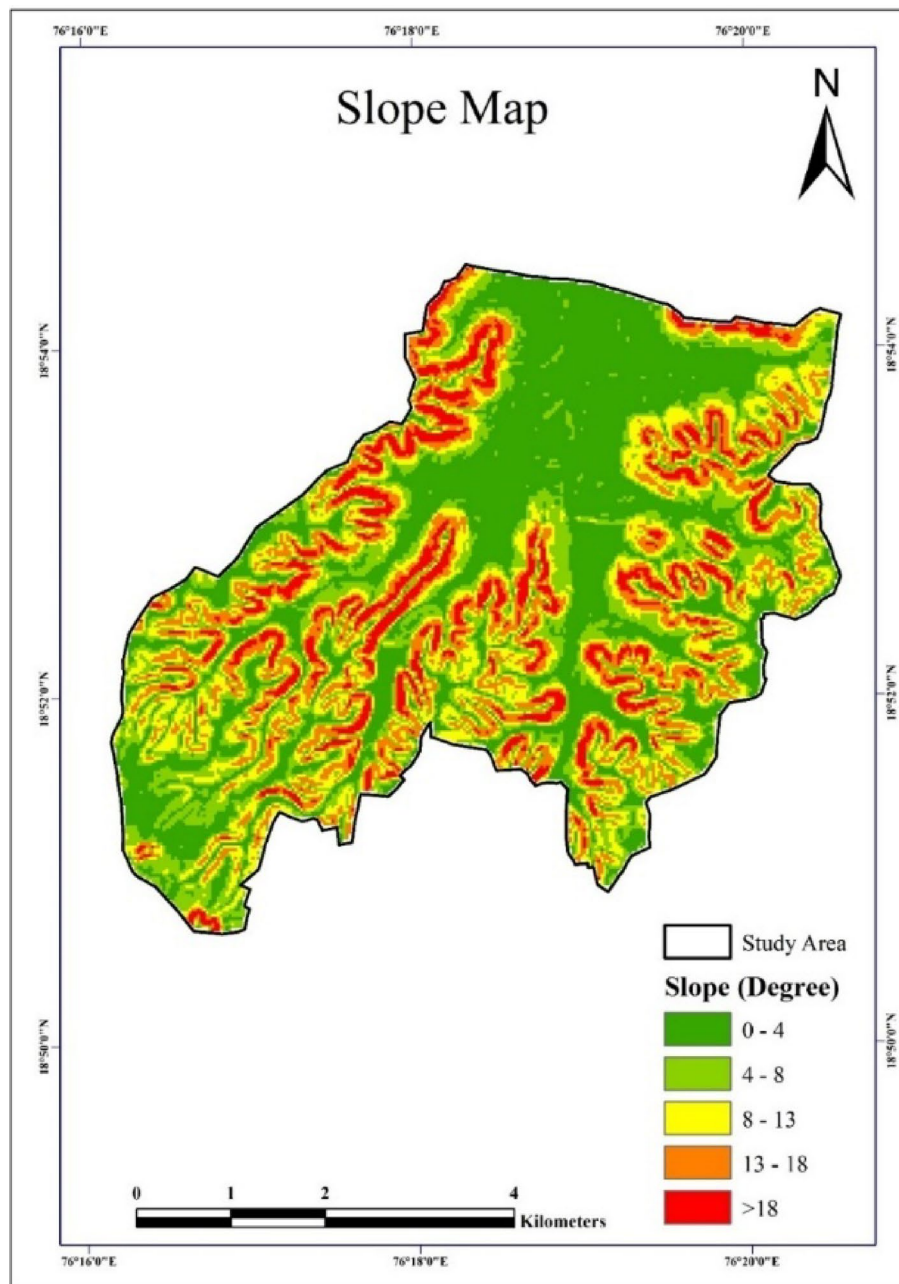


Fig. 6 Slope map of the Moha-Karewadi area

higher ranks because they allow greater residence time and possible weathered-zone development. High-elevation compact basalt units were ranked lower where they coincide with steep slopes and reduced infiltration opportunity. The lithological distribution used for ranking is shown in Fig. 9.

4.7 Land use and land cover

LULC modifies infiltration, runoff and recharge through vegetation cover, irrigation return flow, bare soil exposure and settlement-related compaction. Fallow land is dominant (54.6%), followed by irrigation (16.7%), vegetation (14.3%), wasteland (12.1%), water bodies (1.5%) and settlement (0.8%). Irrigated land and water bodies were ranked

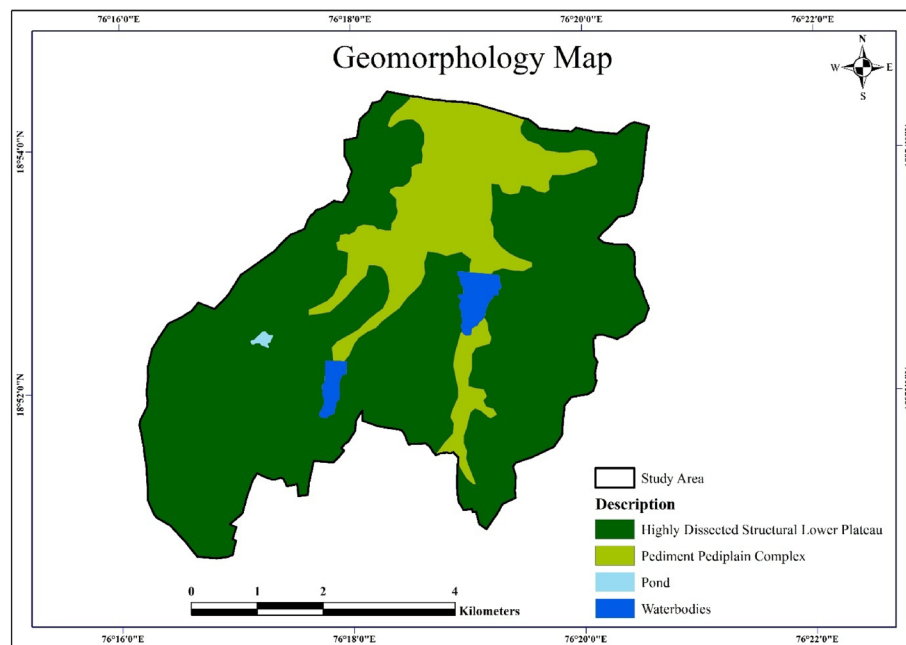


Fig. 7 Geomorphology map showing structural plateau, pediment-pediplain complex, ponds and water bodies

relatively high because they favour local water availability and possible recharge, whereas wasteland was ranked lower due to limited soil cover and reduced residence time. The class-wise LULC distribution is summarised in Table 3. The 2022 LULC classification is shown in Fig. 10.

4.8 Infiltration-rate distribution and recharge relevance

Infiltration rate varied markedly across the study area. The lowest class (4.01–28.68 mm/h) represents slow infiltration and was ranked lowest, whereas the highest class (97.90–178.77 mm/h) represents the most favourable near-surface recharge acceptance and was ranked highest. Field observations indicate that low infiltration is associated with waterlogging, clayey or silty surface layers and incomplete desiltation, whereas higher rates occur in more permeable pediment-pediplain settings. The interpretation therefore emphasises that high infiltration improves recharge suitability only where the subsurface can also accept and transmit water. The spatial distribution of interpolated infiltration rate is shown in Fig. 11.

4.9 VES-based subsurface interpretation

The VES results provide subsurface constraints that are not available from surface GIS layers alone. Around Karewadi, pseudo-section 1 shows a shallow weathered/fractured vesicular-amygdaloidal basalt zone up to about 7 m bgl and a more resistive compact basalt interval at about 8–30 m bgl. A deeper fractured/permeable zone is indicated locally between about 10 and 50 m bgl toward the northeastern part of the section. Pseudo-section 2, which runs parallel to the dam wall, similarly indicates weathered basalt in the upper approximately 7 m and compact basalt below about 8–32 m bgl. The Karewadi pseudo-sections and geoelectric sections are shown in Figs. 12 and 13.

Around Moha, three pseudo-sections show shallow weathered basalt above compact basalt. Section 1 indicates weathered basalt to about 10 m bgl, underlain by a

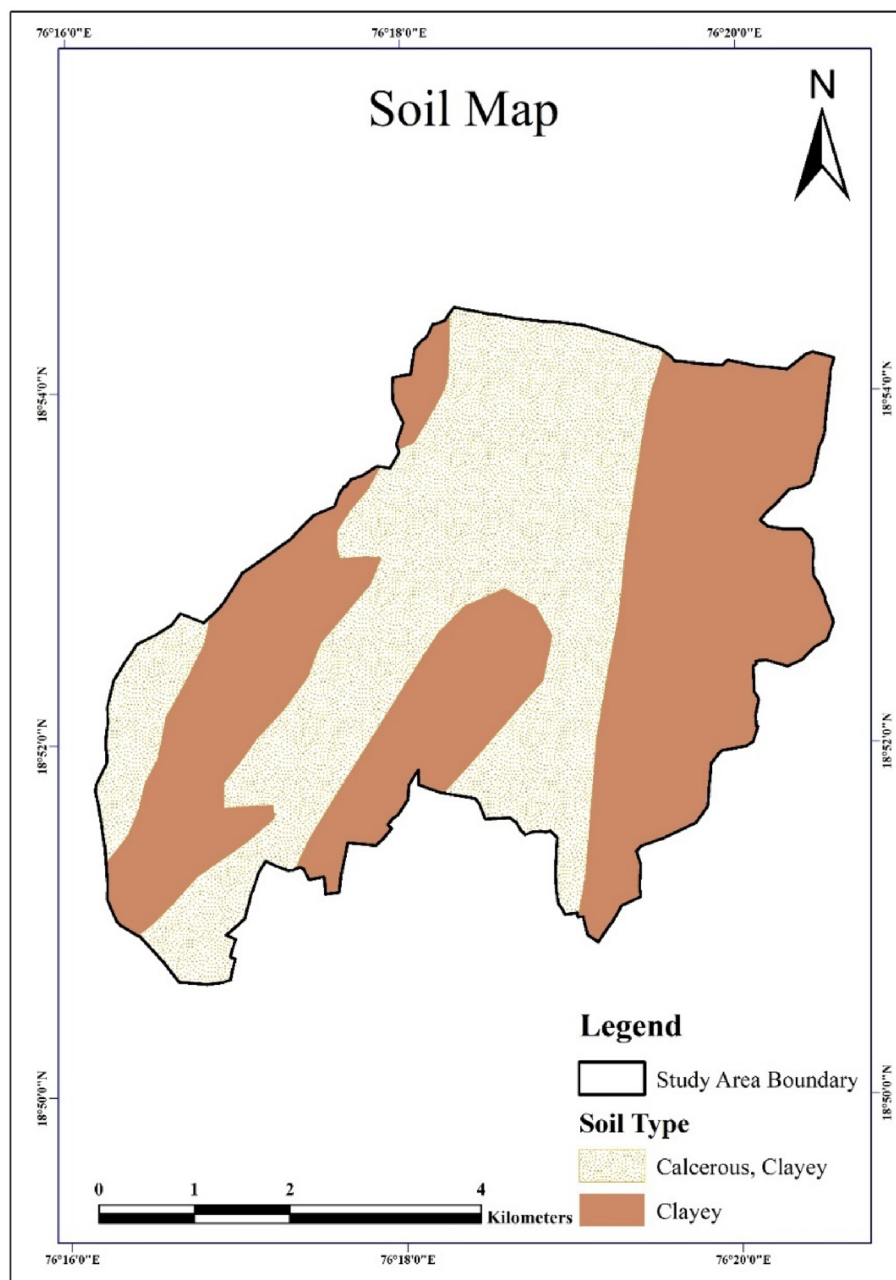


Fig. 8 Soil map of the study area

more resistive compact formation. Section 2 shows higher shallow resistivity toward the northeast and compact basalt at about 10–27 m bgl. Section 3 supports a similar shallow-to-compact basalt configuration. These patterns suggest that recharge benefits may be concentrated in shallow weathered/fractured zones and that compact basalt can restrict deeper percolation. The Moha pseudo-sections and geoelectric sections are shown in Figs. 14, 15 and 16, and the section-wise hydrogeological interpretation is summarised in Table 4.

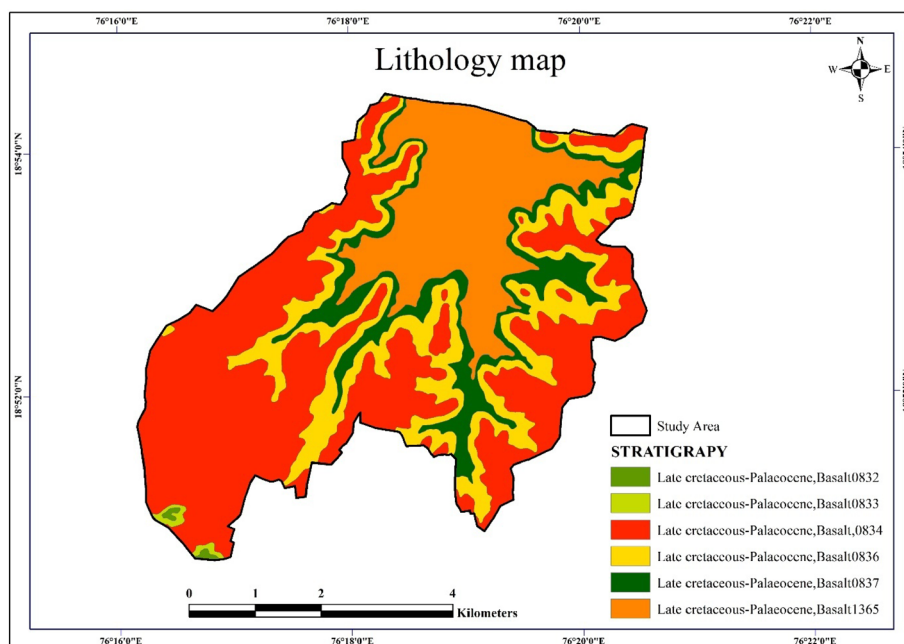


Fig. 9 Lithology map of the study area

Table 3 LULC area distribution in the study area

Sr. No.	Class	Area (%)
1	Water bodies	1.5
2	Irrigation	16.7
3	Fallow land	54.6
4	Settlement	0.8
5	Wasteland	12.1
6	Vegetation	14.3

4.10 Groundwater potential zone map

The GWPZ output was classified into very high, high, moderate, low and very low zones. The very high class covers about 2.07% of the area, and the high class covers about 8.28%. These favourable zones are mainly associated with low-lying geomorphic positions, lower drainage density, higher lineament density and suitable lithology. The limited spatial extent of the very high class is hydrogeologically reasonable for a dissected basaltic terrain, where productive zones are controlled by localised weathering and fracture networks. The final GWPZ map is shown in Fig. 17.

4.11 Groundwater recharge zone map

The GWRZ output includes the infiltration-rate layer and therefore answers a different question: where can water enter the subsurface most effectively? The very high recharge-suitability class covers about 8.86% of the area. Low and very low recharge classes together dominate the landscape, reflecting steep or dissected terrain, low infiltration or unfavourable near-surface conditions. The contrast between the GWPZ and GWRZ maps confirms that groundwater occurrence and recharge suitability should not be treated as identical processes in basaltic terrain. The final GWRZ map is shown in Fig. 18.

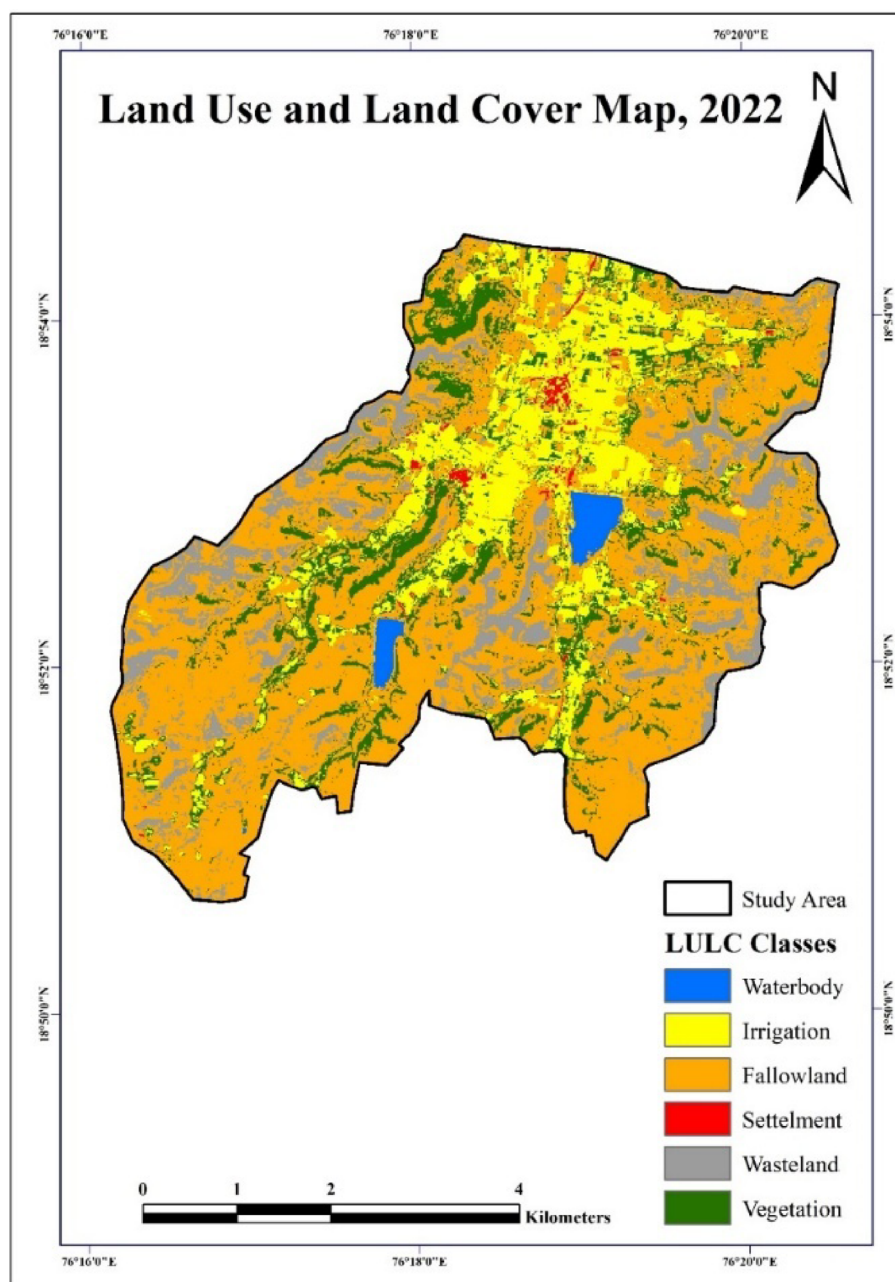


Fig. 10 Land use and land cover map for 2022

4.12 Integrated validation and hydrogeological implications

The validation discussion compares the two maps with VES evidence. In the Karewadi downstream sector, VES results show shallow weathered/fractured basalt and a deeper fractured interval at selected locations, supporting moderate to high GWPZ classes where lineaments and favourable geomorphology coincide. In Moha, high infiltration zones near tank-associated surfaces indicate recharge opportunity, but VES sections also show compact basalt below shallow weathered material; therefore, recharge interventions should be designed to enhance shallow storage and avoid assuming unlimited deep percolation.

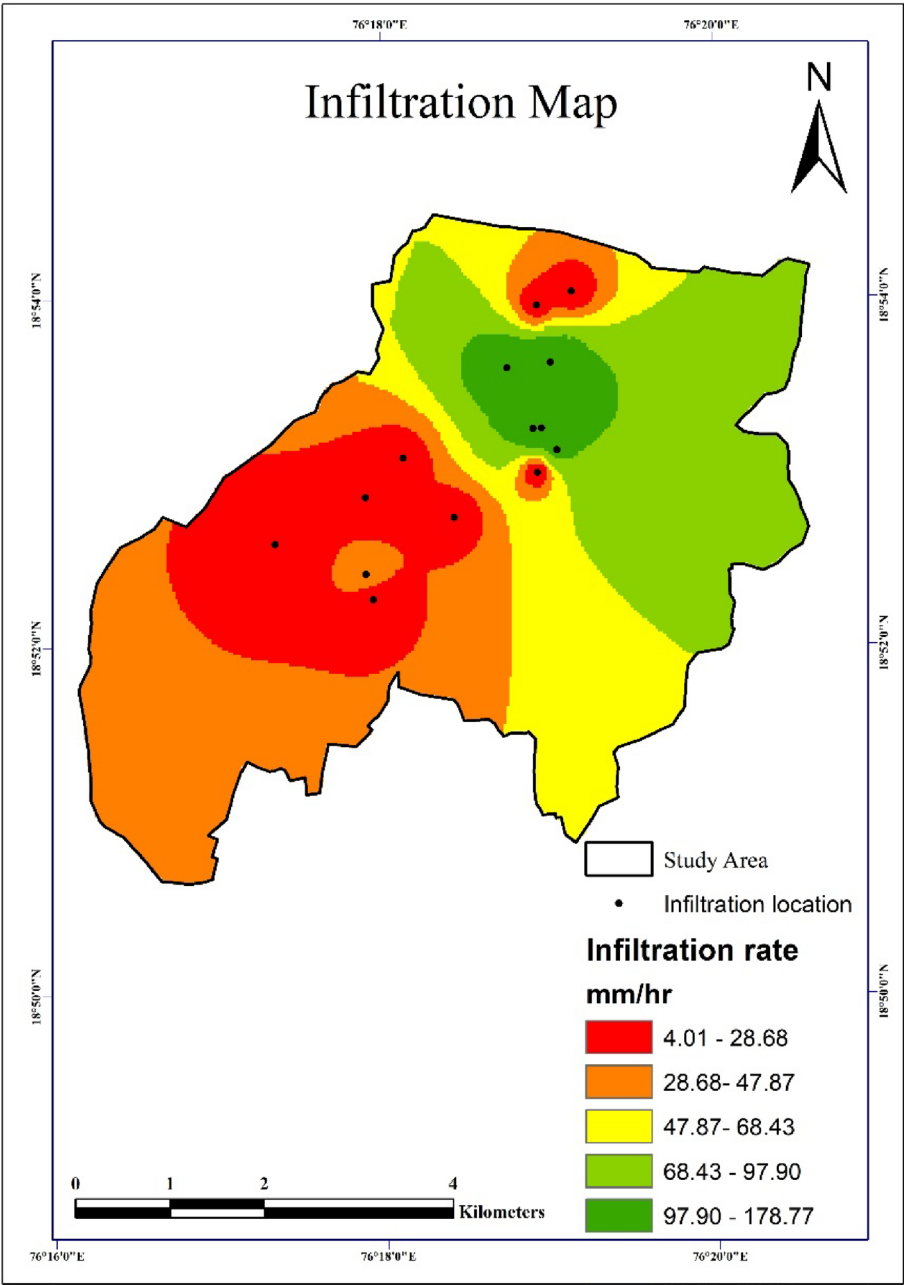


Fig. 11 Infiltration-rate map based on Moha-Karewadi site-level infiltration records retained for the study area

The practical implication is that artificial recharge should not be placed only on the basis of surface AHP suitability. Priority should be given to locations where (i) infiltration rate is moderate to high, (ii) slope and drainage density favour residence time, (iii) lineaments or weathered zones provide storage/transmissivity and (iv) compact basalt barriers do not isolate the recharge zone from the usable aquifer. Low-infiltration areas may still be improved through desiltation, surface treatment, farm ponds or check structures, but they require site-specific design rather than blanket classification as favourable. The class-wise comparison of the two outputs is presented in Fig. 19; Table 5.

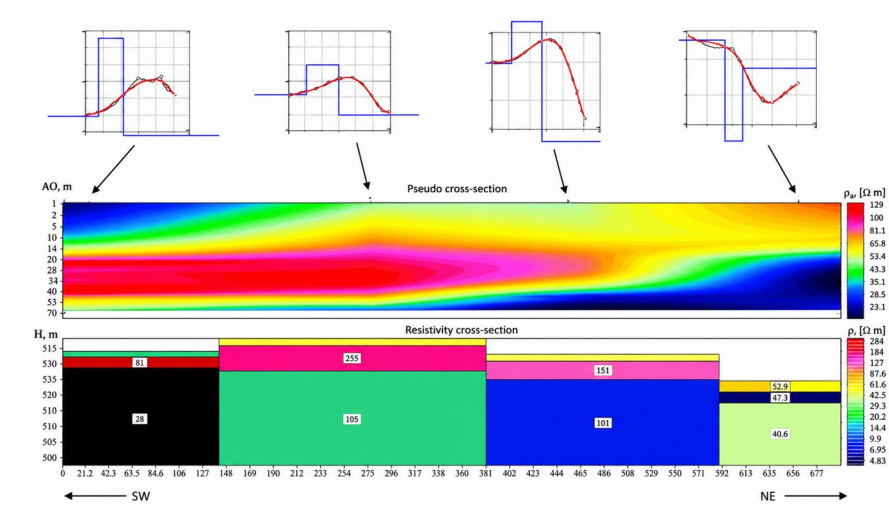


Fig. 12 Karewadi pseudo-section and geoelectric section 1

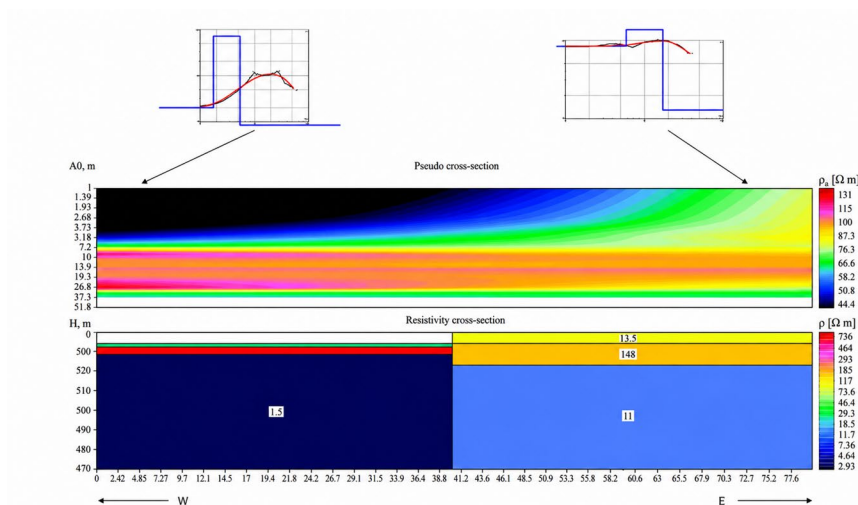


Fig. 13 Karewadi pseudo-section and geoelectric section 2

5 Conclusions

1. The study demonstrates that groundwater potential and recharge suitability should be mapped as related but separate hydrogeological decisions in semi-arid basaltic terrain.
2. The GWPZ map identifies a spatially restricted very high potential class (about 2.07%) and a high class (about 8.28%), mainly where favourable geomorphology, low drainage density, higher lineament density and suitable lithology coincide.
3. The GWRZ map, which adds measured infiltration rate, identifies about 8.86% of the area as very high recharge suitability. This difference confirms that surface potential and actual recharge acceptance are not equivalent.
4. VES interpretation supports the mapped outputs by showing shallow weathered/fractured basalt zones near the tanks, commonly within about 7–10 m bgl, and compact basalt at greater depths in several sections.

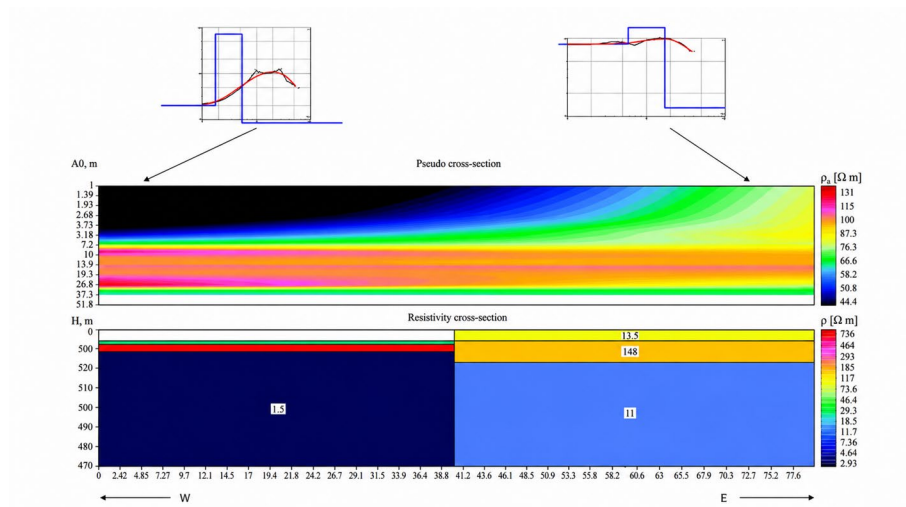


Fig. 14 Moha pseudo-section and geoelectric section 1

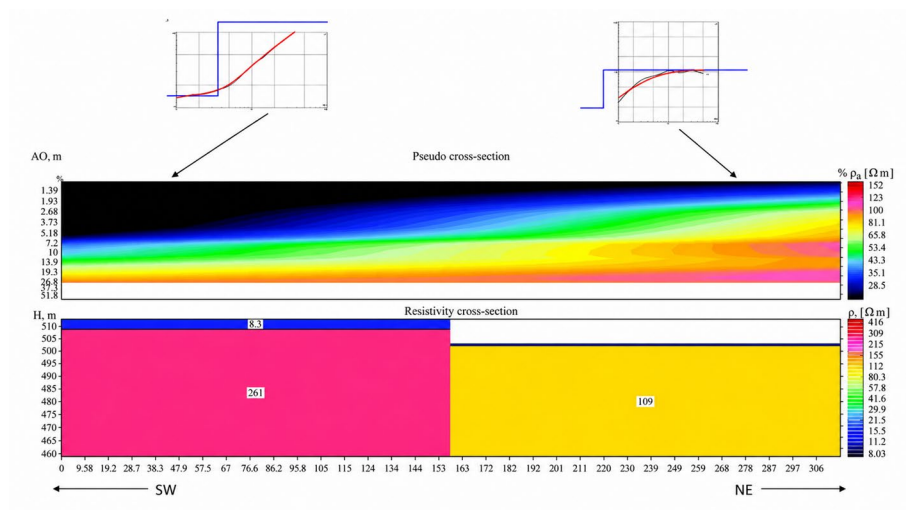


Fig. 15 Moha pseudo-section and geoelectric section 2

5. For recharge planning, priority should be assigned to locations where favourable map classes, measured infiltration and VES-derived weathered/fractured zones overlap. Low-infiltration areas require desiltation or engineered surface treatment before recharge structures are implemented.
6. The workflow is suitable as a first-stage screening tool for data-limited hard-rock watersheds, but final site selection should include site-wise infiltration data, well-yield observations and repeated water-level monitoring where available.

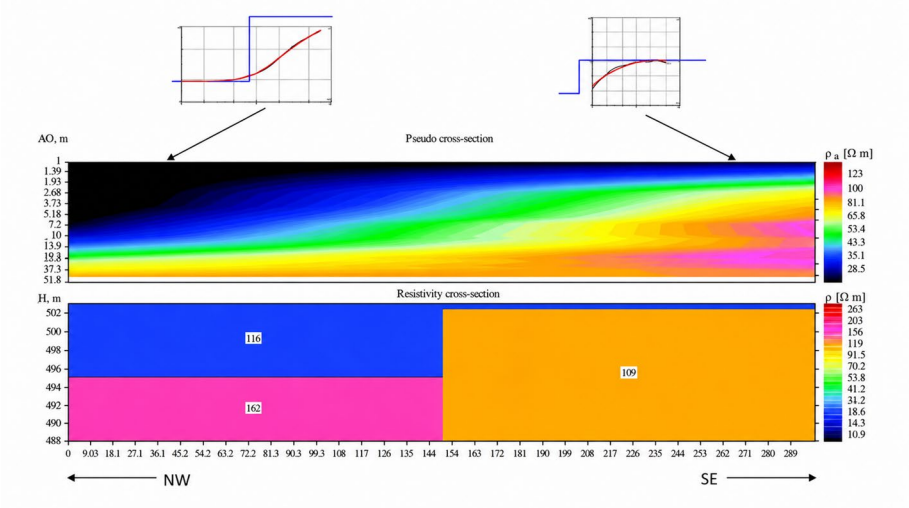


Fig. 16 Moha pseudo-section and geoelectric section 3

Table 4 VES interpretation summary used for field validation

Section	Traverse and length	Main interpreted subsurface units	Recharge/groundwater implication
Karewadi section 1	SW-NE; about 698 m; from tank wall to downstream side	Weathered/fractured vesicular-amygdaloidal basalt to about 7 m bgl; compact basalt at about 8–30 m bgl; local deeper fractured/permeable zone at about 10–50 m bgl toward the NE side	Shallow recharge can support downstream dug wells where weathered/fractured basalt is present; compact basalt limits deeper vertical percolation in parts of the section.
Karewadi section 2	W-E; about 79 m; parallel to tank wall	Weathered basalt to about 7 m bgl; compact basalt at about 8–32 m bgl; possible vesicular-amygdaloidal basalt below about 32 m bgl	Recharge is mainly shallow, with possible deeper storage where less massive basalt or vesicular zones occur.
Moha section 1	S-N; about 484 m; downstream of dam	Weathered basalt to about 10 m bgl; less-weathered to compact basalt below about 10 m bgl	Natural recharge is expected mainly within the shallow weathered horizon.
Moha section 2	SW-NE; about 315 m; downstream of dam	Higher shallow resistivity toward NE; compact basalt at about 10–27 m bgl	Recharge planning should favour shallow/weathered portions rather than areas dominated by compact basalt.
Moha section 3	NW-SE; about 298 m; upstream of dam	Similar shallow-to-compact basalt configuration to Moha section 2	Supports the interpretation that recharge potential is locally controlled by the thickness and continuity of weathered/fractured basalt.

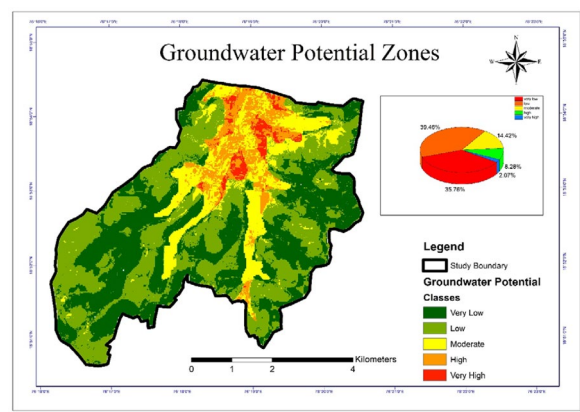


Fig. 17 Groundwater Potential Zone map derived from seven AHP-weighted thematic layers

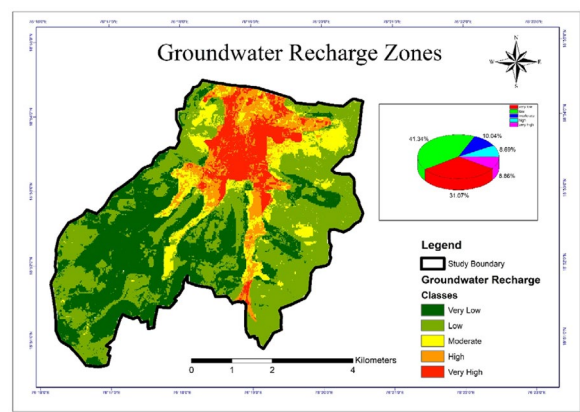


Fig. 18 Groundwater Recharge Zone map generated after adding the infiltration-rate layer

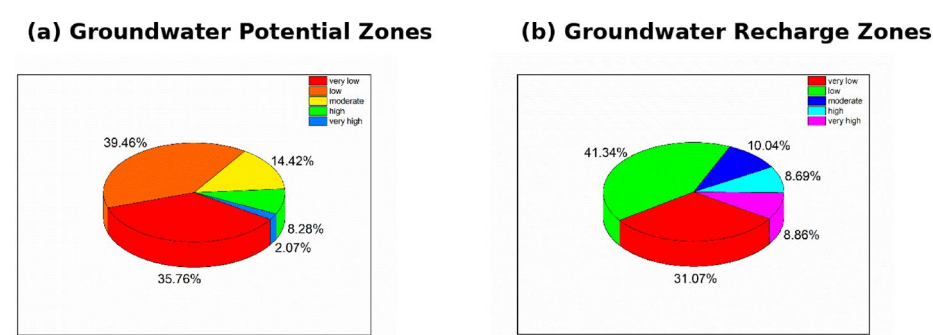


Fig. 19 Area distribution of Groundwater Potential Zone and Groundwater Recharge Zone classes

Class	GWPZ area (%)	GWRZ area (%)
Very high	2.07	8.86
High	8.28	8.69
Moderate	14.42	10.04
Low	39.46	41.34
Very low	35.76	31.07

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1007/s44288-026-00756-3>.

Supplementary Material 1. Supplementary File S1 accompanies the manuscript and contains the supporting AHP, mapped-area and infiltration-data audit tables, together with the ANOVA model decision note.

Acknowledgements

The authors sincerely thank all field staff for their assistance during the fieldwork and gratefully acknowledge the support of the researchers at the WOTR Centre for Resilience Studies (W-CReS), Watershed Organisation Trust (WOTR), Pune, Maharashtra, India.

Author contributions

T.W. conceptualised the study, coordinated the field investigation, collected and interpreted the infiltration and VES data, and contributed to the hydrogeological interpretation. S.S.S. prepared the GIS/remote-sensing layers, carried out AHP-based mapping, generated thematic maps and assisted with spatial analysis. G.B. contributed to the scientific framing, methodology refinement, validation interpretation, manuscript writing, revision and response to reviewers. T.W. and G.B. wrote and revised the main manuscript text. All authors reviewed, edited and approved the final manuscript.

Funding

This research received no specific grant from any funding agency in the public, commercial or not-for-profit sectors.

Data availability

The study used and generated research data, including GIS-derived thematic layers, AHP weighting and mapped-area outputs, field infiltration records, VES-derived geoelectrical interpretations and supporting validation summaries. The key supporting data are provided in Supplementary File S1. Additional raw field sheets, GIS layers and VES interpretation files are available from the corresponding authors upon reasonable request.

Declarations

Ethics approval and consent to participate

Not applicable. The study did not involve human participants, human tissue, animals or clinical data.

Consent for publication

Not applicable. The manuscript contains no identifiable personal information.

Competing interests

The authors declare no competing interests.

Received: 4 March 2026 / Accepted: 21 September 2026

Published online: 29 September 2026

References

1. Amponsah TY, Danuor SK, Wemegah DD, Forson ED. Groundwater potential characterisation over the Voltaian Basin using geophysical, geological, hydrological and topographical datasets. *J Afr Earth Sci.* 2022;192:104558.
2. Aretouyap Z, Asfahani J, Abdulrazzaq ZT, Tchato SC. Contribution of the fuzzy algebraic model to the sustainable management of groundwater resources in the Adamawa watershed. *J Hydrol Reg Stud.* 2022;43:101198.
3. Arshad M, Sarah S, Chatterjee A, Venkatarao A, Ahmed S. Integrated approach to delineate sites for implementation of managed aquifer recharge structures in fluoridated crystalline aquifer of south India. *J Earth Syst Sci.* 2022;131:67.
4. Bera A, Mukhopadhyay BP, Barua S. Delineation of groundwater potential zones in Karha River Basin, Maharashtra, India, using AHP and geospatial techniques. *Arab J Geosci.* 2020;13:1–21.
5. Biswas G, Arshad M, Saba NU, Arora T, Ahmed S. Hydrogeochemical investigation and groundwater quality assessment toward smart city planning in a coastal aquifer, India. *Water Pract Technol.* 2023;18(1):168–84.
6. Biswas G, Arshad M, Mitran T, et al. Electrical resistivity tomography reveals groundwater potential around Lonar Crater, India, in semi-arid basalt terrain. *Discov Geosci.* 2026;4:214. <https://doi.org/10.1007/s44288-026-00597-0>.
7. Biswas G, Arshad M, Arora T, et al. Integrated geospatial modelling and preliminary geophysical appraisal for artificial recharge site selection in the upper tapi basaltic aquifer system, India. *Arab J Geosci.* 2026. <https://doi.org/10.1007/s12517-026-12577-2>.
8. Chatterjee S, Dutta S. Assessment of groundwater potential zone for sustainable water resource management in south-western part of Birbhum District, West Bengal. *Appl Water Sci.* 2022;12:40.
9. Dingman SL. *Physical Hydrology*. 3rd ed. Long Grove, Illinois: Waveland Press; 2015.
10. Echogdali FZ, Boutaleb S, Bendarma A, Saidi ME, Aadraoui M, Abioui M, et al. Application of analytical hierarchy process and geophysical method for groundwater potential mapping in the Tata Basin, Morocco. *Water.* 2022;14(15):2393.
11. Garai T, Biswas G, Santra U. A novel MCDM method based on possibility mean and its application to water resource management problem under bipolar fuzzy environment. In: *Intelligent and Fuzzy Systems: Digital Acceleration and the New Normal*. Cham: Springer; 2022. p. 405–12.
12. Gasmu JM, Rahardjo H, Leong EC. Infiltration effects on stability of a residual soil slope. *Comput Geotech.* 2000;26(2):145–65.
13. Green WH, Ampt GA. Studies on soil physics. *J Agric Sci.* 1911;4(1):1–24.
14. Hussein AA, Govindu V, Nigusse AGM. Evaluation of groundwater potential using geospatial techniques. *Appl Water Sci.* 2017;7:2447–61.

15. Kulkarni H, Deolankar SB, Lalwani A, Joseph B, Pawar S. Hydrogeological framework of the Deccan basalt groundwater systems, west-central India. *Hydrogeol J*. 2000;8(4):368–78. <https://doi.org/10.1007/s100400000079>.
16. Mallick J, Khan RA, Ahmed M, Alqadhi SD, Alsubih M, Falqi I, Hasan MA. Modeling groundwater potential zone in a semi-arid region of Aseer using fuzzy-AHP and geoinformation techniques. *Water*. 2019;11(12):2656.
17. Ogden FL, Raj Pradhan N, Downer CW, Zahner JA. Relative importance of impervious area, drainage density, width function, and subsurface storm drainage on flood runoff from an urbanized catchment. *Water Resour Res*. 2011. <https://doi.org/10.1029/2011WR010550>.
18. Pande CB, Moharir KN, Panneerselvam B, Singh SK, Elbeltagi A, Pham QB, Varade AM, Rajesh J. Delineation of groundwater potential zones for sustainable development and planning using analytical hierarchy process and MIF techniques. *Appl Water Sci*. 2021;11:186.
19. Pathak DR, Hiratsuka A, Awata I, Chen L. Groundwater vulnerability assessment in shallow aquifer of Kathmandu Valley using GIS-based DRASTIC model. *Environ Geol*. 2009;57:1569–78.
20. Rajesh J, Pande CB, Kadam SA, Gorantiwar SD, Shinde MG. Exploration of groundwater potential zones using analytical hierarchical process approach in the Godavari River Basin of Maharashtra, India. *Appl Water Sci*. 2021;11:182.
21. Rao PVN, Appa Rao S, Subba Rao N. Delineation of groundwater prospective zones from a delta region of India, using geoelectrical and water quality approach. *Environ Earth Sci*. 2018;77:616. <https://doi.org/10.1007/s12665-018-7786-7>.
22. Rao PV, Subrahmanyam M, Raju BA. Groundwater exploration in hard-rock terrains of East Godavari district, Andhra Pradesh, India using AHP and WIO analyses together with geoelectrical surveys. *AIMS Geosci*. 2021;7(2):244–67.
23. Sahoo S, Jha MK, Kumar N, Chowdary VM. Evaluation of GIS-based multicriteria decision analysis and probabilistic modeling for exploring groundwater prospects. *Environ Earth Sci*. 2015;74:2223–46.
24. Saaty TL. What is the analytic hierarchy process? In: *Mathematical Models for Decision Support*. Berlin, Heidelberg: Springer; 1988. p. 109–21.
25. Saaty TL. Fundamentals of the analytic hierarchy process. In: *The Analytic Hierarchy Process in Natural Resource and Environmental Decision Making*. Springer; 2001. p. 15–35.
26. Sander P. Lineaments in groundwater exploration: a review of applications and limitations. *Hydrogeol J*. 2007;15(1):71–4.
27. Sonkamble S, Chandra S, Pujari PR. Application of airborne and ground geophysics to unravel the hydrogeological complexity of the Deccan basalts in central India. *Hydrogeol J*. 2022;30:2097–116. <https://doi.org/10.1007/s10040-022-02503-7>.
28. Subba Rao N. Groundwater prospecting and management in an agro-based rural environment of crystalline terrain of India. *Environ Geol*. 2003;43:419–31. <https://doi.org/10.1007/s00254-002-0659-z>.
29. Subba Rao N. Groundwater potential index in a crystalline terrain using remote sensing data. *Environ Geol*. 2006;50(7):1067–76. <https://doi.org/10.1007/s00254-006-0280-7>.
30. Subba Rao N. A numerical scheme for groundwater development in a watershed basin of basement terrain: a case study from India. *Hydrogeol J*. 2009;17:379–96. <https://doi.org/10.1007/s10040-008-0402-2>.
31. Subba Rao N. PI-G: a numerical index for dissemination of groundwater contamination zones. *Hydrol Process*. 2011;26:3344–50. <https://doi.org/10.1002/hyp.8456>.
32. Subba Rao N. Indicators for occurrence of groundwater in the rocks of Eastern Ghats. *Curr Sci*. 2012;103(4):352–3. <https://www.jstor.org/stable/24085075>.
33. Subba Rao N, Gugulothu S, Das R. Deciphering artificial groundwater recharge suitability zones in the agricultural area of a river basin in Andhra Pradesh, India using geospatial techniques and analytical hierarchical process method. *CATENA*. 2022;212:106085. <https://doi.org/10.1016/j.catena.2022.106085>.
34. Sun J, Hu L, Li D, Sun K, Yang Z. Data-driven models for accurate groundwater level prediction and their practical significance in groundwater management. *J Hydrol*. 2022;608:127630.
35. Surinaidu L, Rahman A, Ahmed S. Distributed groundwater recharge potentials assessment based on GIS model and its dynamics in the crystalline rocks of South India. *Sci Rep*. 2021;11:1–16.
36. Umar M, Khan SN, Arshad A, Aslam RA, Khan HMS, Rashid H, Pham QB, Nasir A, Noor R, Khedher KM, Anh DT. A modified approach to quantify aquifer vulnerability to pollution towards sustainable groundwater management in irrigated Indus Basin. *Environ Sci Pollut Res*. 2022;29:1–22.
37. Warsi T, Kumar VS, Dhakate R, Manikyamba C, Rao TV, Rangarajan R. An integrated study of electrical resistivity tomography and infiltration method in deciphering the characteristics and potentiality of unsaturated zone in crystalline rock. *HydroResearch*. 2019;2:109–18.
38. Warsi T, Kumar VS, Kumar D, Nandan MJ, Biswas G, Kumar Sahadevan D, Manikyamba C, Rao TV, Rangarajan R, Ahmed S, Chandrasekhar V. Integration of geophysics and petrography for identifying the aquifer and the rock type: a case study from Giddalur, Andhra Pradesh, India. *J Earth Syst Sci*. 2020;129:1–13.
39. Warsi T, Kulkarni P, Shinde A, Biswas G, D'Souza M. Subsurface filtering of recharge-priority maps in Deccan basalt aquifers: VES-constrained GIS-AHP evidence from Kangti, Telangana, India. *Arab J Geosci*. 2026;19(7):131.
40. Younger PL. *Groundwater in the Environment: An Introduction*. Wiley-Blackwell; 2009.
41. Zhan TLT, Ng CWW. Analytical analysis of rainfall infiltration mechanism in unsaturated soils. *Int J Geomech*. 2004;4(4):273–84.

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.